

National University of Computer & Emerging Sciences

Homework # 2(a)

Counting

1. From a group of n people, suppose that we want to choose a committee of k , $k \leq n$, one of whom is to be designated as chairperson.
 - a. By focusing first on the choice of the committee and then on the choice of the chair, argue that there are $(n \ k)k$ possible choices.
choice of the committee: $(n \ k)$
choice of the chair: k
Total possibilities: $(n \ k)k$
 - b. By focusing first on the choice of the non-chair committee members and then on the choice of the chair, argue that there are $(n \ k - 1)(n - k + 1)$ possible choices.
non-chair committee members: $(n \ k - 1)$ choice of the chair from the remaining people: $(n - k + 1)$
Total possibilities: $(n \ k - 1)(n - k + 1)$
 - c. By focusing first on the choice of the chair and then on the choice of the other committee members, argue that there are $n(n - 1 \ k - 1)$ possible choices.
choice of the chair from all of n people: n
choice of the other committee members: $(n - 1 \ k - 1)$
Total possibilities: $n(n - 1 \ k - 1)$

Conclude from parts (a), (b) and (c) that

$$(n \ k)k = (n \ k - 1)(n - k + 1) = n(n - 1 \ k - 1)$$

by using the factorial definition.

$$\begin{aligned}(n \ k)k &= (n \ k - 1)(n - k + 1) = n(n - 1 \ k - 1) \frac{n! k}{k! (n - k)!} = \frac{n! (n - k + 1)}{(k - 1)! (n - k + 1)!} \\&= \frac{(n - 1)! n}{(k - 1)! (n - 1 - k + 1)!} \frac{n!}{(k - 1)! (n - k)!} = \frac{n!}{(k - 1)! (n - k)!} \\&= \frac{n!}{(k - 1)! (n - k)!}\end{aligned}$$

2. If a multiple-choice test consists of 5 questions, each with 4 possible answers of which only 1 is correct,
 - a. in how many ways can a student check off one answer to each question?
Each question has 4 possible answers, and there are 5 questions. Total ways to answer: $4^5 = 1024$
 - b. in how many ways can a student check off one answer to each question and get all the answers wrong?
Ways to answer incorrectly for all questions: Since there is only 1 correct answer

per question, there are 3 incorrect choices per question. The number of ways to select an incorrect answer for all 5 questions is: $3^5 = 243$

3.

How many distinct permutations can be made from the letters of the word COLUMNS?

$$7!$$

a. How many of these permutations start with the letter M?

If the first letter is **M**, the remaining letters **C, O, L, U, N, S** (6 letters) can be arranged in:

$$6! = 720$$

4.

a. Determine the coefficient of x^9y^3 in the expansion of $(x + 2y)^{12}$.

Using the Binomial Theorem:

$$(x + 2y)^{12} = \sum_{k=0}^{12} \binom{12}{k} x^k (2y)^{12-k},$$

We need the coefficient of x^9y^3 , which means:

$$k = 9$$

The term is:

$$\binom{12}{9} x^9 (2y)^3 = \frac{12!}{9!3!} x^9 (2^3) y^3 = \frac{(12)(11)(10)(8)}{3!} x^9 y^3 = 1760 x^9 y^3$$

So, the coefficient is 1760.

b. Determine the coefficient of xyz^2 in the expansion of $(x + y + 3z)^4$.

5. Consider the following program segment, where i, j and k are integer variables.

For $i := 1$ to 20 do

For $j := 1$ to i do

For $k := 1$ to j do

print ($i * j + k$)

How many times is the print statement executed in this program segment?

The print statement is inside the innermost loop, meaning it executes for each combination of (i, j, k) .

Let's count the number of executions:

$$\begin{aligned} & \sum_{i=1}^{20} \sum_{j=1}^i \sum_{k=1}^j 1 \\ \Rightarrow & \left(\sum_{i=1}^{20} \left(\sum_{j=1}^i \left(\sum_{k=1}^j 1 \right) \right) \right) = \left(\sum_{i=1}^{20} \left(\sum_{j=1}^i (j) \right) \right) = \left(\sum_{i=1}^{20} \left(\frac{i(i+1)}{2} \right) \right) \\ & = \left(\sum_{i=1}^{20} \frac{1}{2} (i^2 + i) \right) = \frac{1}{2} (\sum_{i=1}^{20} i^2 + \sum_{i=1}^{20} i) \\ & = \frac{1}{2} \left(\frac{20(20+1)}{2} + \frac{20(20+1)(40+1)}{6} \right) = 1540 \end{aligned}$$